

SRS Section 8 — Functional System Requirements

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8.1 How to read this section

Each SRS functional requirement carries the identifier **SFR-nnn** and is a testable system statement derived from BRD functional intent. A single BRD requirement may decompose into several SFRs. Where a source requirement is satisfied only in part, the decomposition separates the satisfied behaviour from the unsatisfied behaviour so that status can be stated honestly per unit of delivery.

Every record states both status labels under the [RAID I-09](#) convention: **Catalogue** is the label held on the BRD or the relevant catalogue page, and **RTM** is the label held in the Requirements Traceability Matrix. Where they differ, both are shown and neither is preferred. **MoSCoW** is the inherited Pass #1 source priority and is never altered by implementation status.

Acceptance methods use the following vocabulary: *Automated unit/integration test, Golden-set evaluation, Role-negative test, Contract/schema test, Functional route test, Accessibility audit, Business UAT, Inspection*. An acceptance method being named does not mean it has been executed.

8.2 SFR index

SFR	Title	Source BRD	MoS CoW	Status — Catalogue / RTM
SFR-001	Typed-tool-only query execution	FR-01	Must	Implemented / Designed-not implemented
SFR-002	Query parameter validation and rejection	FR-01	Must	Implemented / Designed-not

				implemented
SFR-003	Result envelope stamping	FR-01, FR-12	Must	Implemented / Partly implemented
SFR-004	Deterministic compliance verdict computation	FR-02	Must	Implemented / Designed-not implemented
SFR-005	Rules-version binding on every verdict	FR-02	Must	Implemented / Designed-not implemented
SFR-006	Rejection of agent override of deterministic logic	FR-02, FR-05	Must	Implemented / Designed-not implemented
SFR-007	Recoverable-denial eligibility filtering	FR-03	Must	Implemented / Designed-not implemented
SFR-008	Recoverable-denial ranking by governed inputs	FR-03	Must	Implemented / Designed-not implemented
SFR-009	Filing-deadline derivation and reconciliation	FR-03	Must	Implemented / Designed-not implemented
SFR-010	Numeric grounding validation before display	FR-04	Must	Implemented / Partly implemented
SFR-011	Fail-closed grounding block state	FR-04	Must	Implemented / Partly implemented
SFR-012	Server-side role capability enforcement	FR-05	Must	Implemented / Designed-not implemented
SFR-013	Aggregate-only viewer data scope	FR-05, BR-02	Must	Implemented / Designed-not implemented

SFR-014	Prompt-escalation resistance	FR-05	Must	Implemented / Designed-not implemented
SFR-015	Chart specification bound to an existing session result	FR-06	Must	Designed-not implemented / Partly implemented
SFR-016	Chart type and field whitelist validation	FR-06	Must	Designed-not implemented / Partly implemented
SFR-017	Executive Insight KPI band	FR-07	Must	Designed-not implemented / Designed-not implemented
SFR-018	Executive Insight governed narrative and chart canvas	FR-07	Must	Designed-not implemented / Designed-not implemented
SFR-019	Executive Insight source affordance	FR-07, FR-12	Must	Designed-not implemented / Designed-not implemented
SFR-020	Analyst Workbench governed query surface	FR-08	Must	Designed-not implemented / Partly implemented
SFR-021	Analyst Workbench denial-queue surface	FR-08, FR-16	Must	Designed-not implemented / Partly implemented
SFR-022	Analyst Workbench appeal-review surface	FR-08, FR-10	Must	Designed-not implemented / Designed-not implemented
SFR-023	Audit Console event search and retrieval	FR-09	Must	Designed-not implemented /

				Designed-not implemented
SFR-024	Rules, model and configuration version history	FR-09	Must	Designed-not implemented / Designed-not implemented
SFR-025	Append-only audit enforcement	FR-09, NFR-05	Must	Designed-not implemented / Designed-not implemented
SFR-026	Governed appeal draft generation	FR-10	Must	Designed-not implemented / Designed-not implemented
SFR-027	Pending-review state; no autonomous approval or send	FR-10, BR-04	Must	Implemented / Designed-not implemented
SFR-028	Prohibited-language and evidence-checklist validation	FR-10	Must	Designed-not implemented / Designed-not implemented
SFR-029	Named-human approval event recording	FR-11	Must	Designed-not implemented / Designed-not implemented
SFR-030	Duplicate approval prevention and immutable retrieval	FR-11	Must	Designed-not implemented / Designed-not implemented
SFR-031	End-to-end evidence lineage resolution	FR-12	Must	Designed-not implemented / Partly implemented

SFR-032	Extract schema, enum and window validation	FR-13	Must	Implemented / Implemented
SFR-033	Generation tagging and artefact/truth regeneration	FR-13	Must	Implemented / Implemented
SFR-034	Dry-run evaluation gate and human acceptance decision	FR-13, BR-06	Must	Implemented / Implemented
SFR-035	Atomic rollback on generation rejection	FR-13, NFR-06	Must	Implemented / Implemented
SFR-036	Read-only agent-plane boundary	FR-14	Must	Implemented / Implemented
SFR-037	Ingestion permission separation	FR-14	Must	Implemented / Implemented
SFR-038	KPI dashboard and filter behaviour	FR-15	Should	Designed-not implemented / Designed-not implemented
SFR-039	Export controls and approved formatting	FR-15, SEC-05	Should	Designed-not implemented / Designed-not implemented
SFR-040	Full recoverable-denials queue behaviour	FR-16	Should	Designed-not implemented / Partly implemented
SFR-041	Insights library with severity, impact and drill-down	FR-17	Should	Designed-not implemented / Designed-not implemented
SFR-042	Profile, preferences and access presentation	FR-18, SEC-04	Should	Designed-not implemented /

				Designed-not implemented
SFR-043	Adaptive single-table intake profiling — future Gen-3	FR-19	Could	Designed-not implemented / Designed-not implemented
SFR-044	Period comparison and change decomposition — future Gen-3	FR-20	Could	Designed-not implemented / Designed-not implemented
SFR-045	Deterministic anomaly detection — future Gen-3	FR-21	Could	Designed-not implemented / Designed-not implemented
SFR-046	Proactive cited findings feed — future Gen-3	FR-22	Could	Designed-not implemented / Designed-not implemented
SFR-047	Exclusion — no autonomous external appeal submission	FR-23	Won't	Deferred / Deferred
SFR-048	Exclusion — no production claims or EHR write	FR-24	Won't	Deferred / Deferred
SFR-049	Exclusion — no automatic multi-table relationship discovery	FR-25	Won't	Deferred / Deferred
SFR-050	Exclusion — no autonomous workers or scheduled cohort scoring	FR-26	Won't	Deferred / Deferred

8.3 Governed query and semantic services

SFR-001 — Typed-tool-only query execution

Shall. The system shall execute every quantitative data request exclusively through a registered typed tool bound to an approved semantic view, and shall provide no path by which free-text SQL originating from a user or from the language model reaches the data layer.

Trace. Source BRD: FR-01, BR-03. Process: P-01, P-03, P-04. Use case: UC-02, UC-04. UI trace: not applicable at this level. Test trace: automated unit and integration tests; golden-set evaluation. Dependencies: DEP-03, DEP-05.

Actor / component. Agent dispatcher and typed semantic tool layer, on behalf of executive/viewer, analyst and admin actors.

Preconditions. An authenticated session with a resolved role; a loaded dataset configuration; an available governed replica. **Inputs.** Tool name and typed parameter set. **Outputs.** A stamped result envelope containing values, query identifier and lineage metadata.

Required behaviour. The dispatcher resolves the requested tool against the registered contract set, validates the parameters against the contract, executes the bound semantic view, and returns a stamped envelope. Any request that does not resolve to a registered contract is refused.

Alternative / error behaviour. An unregistered tool name, an attempt to pass raw SQL as a parameter value, or a request for an unbound view is rejected with a structured error. No partial result is returned and no query is executed.

Security and access rule. Tool availability is constrained by role before execution; the boundary is enforced in code and never by prompt text. **Audit and evidence.** Tool name, parameters, resolved role, query identifier and execution time are recorded for every invocation, including refusals.

Acceptance method. Automated unit and integration test, plus golden-set evaluation of governed query items.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must. Repository verification supports the backend reading; the divergence is the I-09 scope difference described in the parent document.

Open items. Metric and field semantics remain subject to DD-01 to DD-14 under I-06.

SFR-002 — Query parameter validation and rejection

Shall. The system shall validate every tool parameter against its declared contract type, permitted enumeration and permitted range before execution, and shall reject a non-conforming

request without executing it.

Trace. Source BRD: FR-01. Process: P-01, P-03, P-04. Use case: UC-02, UC-04. Test trace: contract/schema test; automated unit test. Dependencies: DEP-03.

Actor / component. Agent dispatcher; tool contract validator.

Preconditions. A registered contract exists for the requested tool. **Inputs.** Candidate parameter set. **Outputs.** Either an executed query result or a structured validation error naming the offending parameter.

Required behaviour. Validation occurs before any data access. Enumerated values are checked against the configuration-resolved enumeration for the active dataset generation, so that a value valid for one generation is not silently accepted for another.

Alternative / error behaviour. A type mismatch, an out-of-range value or an unrecognised enumeration member produces a rejection that identifies the parameter and the expected form, and produces no data access.

Security and access rule. Validation failures do not disclose data values, schema internals or record counts. **Audit and evidence.** The rejected parameter name and reason are recorded; parameter values are recorded subject to the data-classification rules in section 12.

Acceptance method. Contract/schema test with negative cases per parameter type.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Permitted enumerations depend on unresolved data definitions under I-06.

SFR-003 — Result envelope stamping

Shall. The system shall stamp every tool result envelope with a unique query identifier, the executed parameters, the execution timestamp, the dataset generation tag, the configuration version and the configuration hash.

Trace. Source BRD: FR-01, FR-12, BR-05. Process: P-03, P-04, P-05. Use case: UC-02, UC-05, UC-08. UI trace: Source and Evidence Detail. Test trace: automated unit test; inspection of retained envelopes. Dependencies: DEP-03, DEP-05.

Actor / component. Typed tool layer; result store.

Preconditions. An accepted dataset generation with a resolved configuration version and hash.

Inputs. Executed query and its parameters. **Outputs.** A stamped, retrievable result envelope.

Required behaviour. Stamping is applied at production of the envelope and is not reconstructible after the fact. The stamped identifier is the sole key by which a displayed figure

is later reconciled to its source.

Alternative / error behaviour. If configuration version or hash cannot be resolved, the tool fails closed and returns no result rather than returning an unstamped envelope.

Security and access rule. Envelope retrieval is role-gated to the same scope as the originating query. **Audit and evidence.** The envelope is itself the primary evidence artefact for lineage requirements in section 10.

Acceptance method. Automated unit test plus inspection of retained run artefacts.

Status. Catalogue: **Implemented** · RTM: **Partly implemented** · MoSCoW (source): Must. Backend stamping is verified; presentation of the stamp to a user is not implemented.

Open items. Retention period for stamped envelopes is unapproved and is carried as a TBD in section 9 under SNFR-014.

8.4 Deterministic compliance services

SFR-004 — Deterministic compliance verdict computation

Shall. The system shall compute every compliance verdict in versioned deterministic code, and shall not permit the language model to compute, infer, adjust or narrate a verdict that the rules engine did not produce.

Trace. Source BRD: FR-02, BR-03, BR-05. Process: P-04, P-02. Use case: UC-05, UC-06. UI trace: Audit Console; Source and Evidence Detail. Test trace: automated unit test with fixtures; golden-set evaluation. Dependencies: DEP-02, DEP-03, DEP-05.

Actor / component. Deterministic rules engine, invoked through the `check_compliance` tool contract.

Preconditions. A loaded, versioned ruleset; a resolvable claim or claim set. **Inputs.** Claim identifier or claim attributes and the applicable rule scope. **Outputs.** A verdict, the triggering rule references and the ruleset version.

Required behaviour. Identical inputs against an identical ruleset version return an identical verdict on repeat execution. The verdict is produced before any narration and is passed to the assembler as a fixed value.

Alternative / error behaviour. An unresolvable claim, an out-of-scope rule request or an unloaded ruleset returns a structured error. No provisional or estimated verdict is produced.

Security and access rule. Verdict retrieval is role-gated; record-level verdicts are unavailable to the viewer role. **Audit and evidence.** Verdict, rule references, ruleset version and input identifiers are recorded.

Acceptance method. Automated unit test against retained fixtures, repeated-execution determinism test, and golden-set evaluation.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Rule applicability depends on jurisdiction confirmation under DEP-02 and assumption A-03; D-BRD-06 remains Proposed.

SFR-005 — Rules-version binding on every verdict

Shall. The system shall return the identifier of the ruleset version that produced each verdict, and shall retain that identifier with the verdict for later reconstruction.

Trace. Source BRD: FR-02, FR-12. Process: P-04. Use case: UC-05, UC-08. UI trace: Audit Console version history. Test trace: automated unit test. Dependencies: DEP-02, DEP-05.

Actor / component. Rules engine; result store.

Preconditions. A versioned ruleset is loaded. **Inputs.** Verdict computation request. **Outputs.** Verdict plus ruleset version identifier.

Required behaviour. The version is captured at computation time, not at retrieval time, so that a later ruleset change does not alter the recorded provenance of a past verdict.

Alternative / error behaviour. If the ruleset version cannot be resolved, the computation fails closed and no verdict is returned.

Security and access rule. Version identifiers are visible to analyst and admin roles. **Audit and evidence.** The version identifier is a mandatory audit field.

Acceptance method. Automated unit test; inspection of retained verdict records.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. None specific to this requirement.

SFR-006 — Rejection of agent override of deterministic logic

Shall. The system shall reject any attempt, whether originating in user input or in model output, to substitute, override or recompute a deterministic compliance verdict or a governed metric value.

Trace. Source BRD: FR-02, FR-05, BR-05, SEC-09. Process: P-04. Use case: UC-05. Test trace: role-negative test; golden-set safety items. Dependencies: DEP-05.

Actor / component. Agent dispatcher; grounding validator; rules engine boundary.

Preconditions. An active session. **Inputs.** Any instruction requesting recomputation, override or adjustment. **Outputs.** A refusal, and the unmodified deterministic result where one exists.

Required behaviour. The boundary holds regardless of the phrasing, claimed authority or apparent urgency of the instruction. Refusal explains the boundary without disclosing rule internals.

Alternative / error behaviour. Repeated override attempts within a session are recorded and do not weaken the boundary.

Security and access rule. Prompt text is never a security boundary; enforcement is server-side.

Audit and evidence. Override attempts are recorded with session, role and instruction reference.

Acceptance method. Role-negative test and golden-set safety evaluation.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Golden-set item GQ-048 remains unadjudicated under I-11 and may not be cited as evidence for or against this requirement.

8.5 Recoverable-denial services

SFR-007 — Recoverable-denial eligibility filtering

Shall. The system shall return only denials that are both denied and within the applicable resubmission filing window, and shall exclude any denial outside that window from the recoverable set.

Trace. Source BRD: FR-03, BR-03. Process: P-01. Use case: UC-04. UI trace: Recoverable Denials Queue. Test trace: automated unit test; golden-set evaluation. Dependencies: DEP-01, DEP-03, DEP-05.

Actor / component. `get_recoverable_denials` composite tool over governed semantic views.

Preconditions. An accepted dataset generation with a resolved as-of date. **Inputs.** Optional payer, amount-threshold and period filters. **Outputs.** An eligible denial set with per-item recoverability attributes.

Required behaviour. Eligibility is computed against the governed as-of date of the accepted generation, not against wall-clock time, so that results are reproducible for a given generation.

Alternative / error behaviour. An empty eligible set is returned as an explicit empty result with its stamped envelope, never as an error and never as an unfiltered set.

Security and access rule. Record-level denial detail is unavailable to the viewer role; aggregate counts remain available. **Audit and evidence.** Filter parameters, as-of date and result identifier are recorded.

Acceptance method. Automated unit test with boundary cases at the window edge; golden-set evaluation.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. The 90-day filing logic is an unvalidated assumption under A-03 and may not be presented as a contractual position.

SFR-008 — Recoverable-denial ranking by governed inputs

Shall. The system shall rank eligible denials using governed denied amount, filing-window position and overturn-likelihood inputs, and shall derive no ranking factor from language-model judgement.

Trace. Source BRD: FR-03, BR-03. Process: P-01. Use case: UC-04. UI trace: Recoverable Denials Queue. Test trace: automated unit test; golden-set evaluation. Dependencies: DEP-03, DEP-05.

Actor / component. Recoverable-denials composite tool.

Preconditions. An eligible denial set per SFR-007. **Inputs.** Governed amount, window and overturn inputs. **Outputs.** An ordered set with the ranking value exposed per item.

Required behaviour. The ranking value and every input to it are returned with the result so that the order can be reconciled without re-derivation.

Alternative / error behaviour. A missing ranking input for an item excludes that item from ranking and reports it as unrankable rather than assigning a default value.

Security and access rule. As SFR-007. **Audit and evidence.** Ranking inputs and the resulting order are reconstructible from the stamped envelope.

Acceptance method. Automated unit test; reconciliation test of ranking output against governed inputs.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Expected-value semantics remain subject to DD-01 to DD-14 under I-06.

Shall. The system shall derive and expose a filing deadline for every eligible denial, and that deadline shall reconcile to the governed denial date and the applicable window rule.

Trace. Source BRD: FR-03, FR-16. Process: P-01, P-02. Use case: UC-04, UC-06. UI trace: Recoverable Denials Queue. Test trace: automated unit test. Dependencies: DEP-01, DEP-03.

Actor / component. Recoverable-denials composite tool; rules engine window logic.

Preconditions. A governed denial date and an applicable window rule. **Inputs.** Denial record and rule scope. **Outputs.** Deadline date and remaining-days value relative to the governed as-of date.

Required behaviour. Remaining days are computed from the generation as-of date and are labelled as such, so that a stale view cannot present an expired window as open.

Alternative / error behaviour. Where the window rule is unresolved for a payer or contract, the deadline is reported as undetermined rather than defaulted.

Security and access rule. As SFR-007. **Audit and evidence.** Deadline, source date and rule reference are retained with the result.

Acceptance method. Automated unit test with boundary and undetermined cases.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Contract-specific windows depend on DEP-02 and A-03.

8.6 Grounding and response integrity

Shall. The system shall validate every numeric statement in a candidate response against retained trusted tool results before that response is released, and shall release no numeric value that does not trace to such a result.

Trace. Source BRD: FR-04, BR-05, NFR-01. Process: P-03, P-04. Use case: UC-02, UC-03, UC-05. UI trace: all display surfaces. Test trace: automated unit test; golden-set evaluation; retained grounding test set. Dependencies: DEP-03, DEP-05, DEP-08.

Actor / component. Grounding validator within the response assembler.

Preconditions. One or more stamped result envelopes in session scope. **Inputs.** Candidate response text and the session result set. **Outputs.** Either a released response with resolvable citations, or a blocked response.

Required behaviour. Validation is applied to the assembled response, not to intermediate model output alone, so that a value introduced late in assembly is still caught. Derived figures the model computed itself are treated as unsupported even where the inputs were trusted.

Alternative / error behaviour. A bounded re-ask may be attempted; if validation still fails the response is blocked under SFR-011. The number of attempts is bounded and recorded.

Security and access rule. Validation runs server-side and cannot be disabled by session input.

Audit and evidence. Grounding attempts, outcomes and the identifiers of the results relied upon are recorded.

Acceptance method. Automated unit test, retained grounding test set, and golden-set evaluation.

Status. Catalogue: **Implemented** · RTM: **Partly implemented** · MoSCoW (source): Must. Backend validation is verified; no display surface exists to which it is applied end to end.

Open items. The complete post-fix live evidence required to demonstrate this behaviour under live conditions is incomplete under R-02 and DEP-08.

SFR-011 — Fail-closed grounding block state

Shall. On grounding failure the system shall withhold the unsupported content entirely, present an explicit blocked state, and retain a trace identifier that resolves to the blocked output.

Trace. Source BRD: FR-04, NFR-01, UAT-07. Process: P-03, P-04. Use case: UC-02, UC-03. UI trace: grounding-error state on all display surfaces. Test trace: automated unit test; UAT-07. Dependencies: DEP-05, DEP-08.

Actor / component. Response assembler; presentation layer.

Preconditions. A failed grounding validation. **Inputs.** Blocked candidate response. **Outputs.** An accessible blocked-state message and a retained trace identifier.

Required behaviour. The blocked state never displays the unsupported figure, not even as an example of what was blocked. Partial release of a response containing one unsupported figure is not permitted.

Alternative / error behaviour. Where a response contains both supported and unsupported content, the whole response is blocked rather than partially released.

Security and access rule. The blocked-state message discloses no data values. **Audit and evidence.** Blocked outputs are retained and are retrievable by trace identifier.

Acceptance method. Automated unit test with a deliberately unsupported statement; UAT-07 at business acceptance.

Status. Catalogue: **Implemented** · RTM: **Partly implemented** · MoSCoW (source): Must.

Open items. The accessible presentation of the blocked state is unimplemented and is specified for acceptance in section 11 under NFR-15.

8.7 Role and access enforcement

SFR-012 — Server-side role capability enforcement

Shall. The system shall determine tool, route and data capability from the server-resolved role of the session, and shall enforce that determination in code before execution.

Trace. Source BRD: FR-05, NFR-03, SEC-04. Process: P-01, P-03, P-04. Use case: UC-02, UC-04, UC-05, UC-08, UC-09. UI trace: denied state on all surfaces. Test trace: role-negative test matrix. Dependencies: DEP-01, DEP-05, DEP-07.

Actor / component. Agent dispatcher; application API boundary when implemented.

Preconditions. An authenticated session with a resolved role. **Inputs.** Requested capability.

Outputs. Execution or a denial.

Required behaviour. The capability decision precedes execution and is independent of any assertion made in session text about the user's role or authority.

Alternative / error behaviour. Denial returns an explicit unauthorised state that names the required role class without disclosing the withheld data.

Security and access rule. Least privilege by default; prompt text is not a security boundary.

Audit and evidence. Every denial is recorded with session, role, requested capability and time.

Acceptance method. Role-negative test matrix covering every persona against every capability.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must. Dispatcher enforcement is verified; production identity and access management is absent.

Open items. Production authentication is undelivered under DEP-05; role fit is unvalidated under A-02.

SFR-013 — Aggregate-only viewer data scope

Shall. The system shall restrict the executive/viewer role to aggregate results, and shall deny that role any claim-level row, member identifier or record-level export across every tool, route and export path.

Trace. Source BRD: FR-05, BR-02, NFR-04, SEC-05, UAT-02. Process: P-03. Use case: UC-02, UC-03. UI trace: Executive Insight; Full KPI Dashboard; denied state. Test trace: role-negative test; UAT-02. Dependencies: DEP-01, DEP-03, DEP-05, DEP-07.

Actor / component. Dispatcher; semantic view layer; export controls when implemented.

Preconditions. A session resolved to the viewer role. **Inputs.** Any request that would return record-level data. **Outputs.** Aggregate results, or a denial.

Required behaviour. The restriction is applied at the data layer, not by filtering a record-level result after retrieval, so that record-level data is never materialised for a viewer session.

Alternative / error behaviour. A viewer request for record-level detail returns a denial that offers the available aggregate alternatives.

Security and access rule. Aggregate-only is a privacy control, not a presentation preference.

Audit and evidence. Denials are recorded; aggregate results carry their normal lineage stamp.

Acceptance method. Role-negative test across tool, route and export paths; UAT-02 at business acceptance.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must. Backend boundary verified; UI and export paths do not exist and are therefore untested.

Open items. Export-path enforcement cannot be evidenced until export exists under DEP-05.

SFR-014 — Prompt-escalation resistance

Shall. The system shall maintain its role, data and safety boundaries when session input asserts a privileged mode, claims administrative authority, or instructs the system to disregard its instructions.

Trace. Source BRD: FR-05, NFR-03, SEC-04. Process: P-03. Use case: UC-02. Test trace: golden-set safety items; role-negative test. Dependencies: DEP-05.

Actor / component. Dispatcher; system prompt boundary; grounding validator.

Preconditions. An active session of any role. **Inputs.** Escalation or instruction-override attempt.

Outputs. A refusal that preserves the boundary.

Required behaviour. Refusal states that the requested access is role-gated and does not reveal system prompt contents, withheld records or the existence of a privileged mode.

Alternative / error behaviour. The session continues normally after refusal; no punitive termination is required.

Security and access rule. Enforcement is server-side and independent of prompt phrasing.

Audit and evidence. The attempt, the refusal and the session context are recorded.

Acceptance method. Golden-set safety evaluation and role-negative test.

Status. Catalogue: **Implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. The principal evaluation item for this behaviour, GQ-048, is unadjudicated under I-11 and may not be counted as either a pass or a failure.

8.8 Chart contract services

SFR-015 — Chart specification bound to an existing session result

Shall. The system shall produce a chart specification only from a result already present in session scope and identified by its stamped query identifier, and shall not accept literal data values in a chart request.

Trace. Source BRD: FR-06, BR-05. Process: P-03. Use case: UC-02, UC-03. UI trace: Executive Insight chart canvas; Full KPI Dashboard. Test trace: automated unit test; golden-set chart items. Dependencies: DEP-05, DEP-07.

Actor / component. `render_chart` tool; response assembler.

Preconditions. A stamped result envelope in session scope. **Inputs.** Query identifier, chart type and field selection. **Outputs.** A validated chart specification referencing the source result.

Required behaviour. The specification carries the originating query identifier so that a rendered chart is reconcilable to its source without re-query.

Alternative / error behaviour. An unknown query identifier, an expired session result or an attempt to supply inline values is rejected and no specification is produced.

Security and access rule. The chart inherits the access scope of its source result. **Audit and evidence.** Specification, source identifier and field selection are recorded.

Acceptance method. Automated unit test; golden-set chart-specification items.

Status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented** · MoSCoW (source): Must. The backend contract is implemented; no renderer exists.

Open items. Golden-set item GQ-038 exercises this path and has never passed a live run; it is the open evidence gap under R-02 and DEP-08.

Shall. The system shall accept only whitelisted chart types and only fields present in the referenced result, and shall reject any other request.

Trace. Source BRD: FR-06. Process: P-03. Use case: UC-02, UC-03. UI trace: chart error state. Test trace: contract/schema test. Dependencies: DEP-05.

Actor / component. `render_chart` contract validator.

Preconditions. A referenced result envelope. **Inputs.** Requested chart type and fields. **Outputs.** A validated specification or a structured rejection.

Required behaviour. Field validity is checked against the referenced result's actual schema rather than against an expected schema, so that a generation change cannot silently produce an invalid chart.

Alternative / error behaviour. Rejection names the unsupported type or absent field and displays no partial chart.

Security and access rule. As SFR-015. **Audit and evidence.** Rejections are recorded with reason.

Acceptance method. Contract/schema test with negative cases per chart type.

Status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented** · MoSCoW (source): Must.

Open items. The approved chart-type whitelist is design evidence only pending UI implementation.

8.9 Executive Insight surface

 SFR-017 to SFR-025 specify user-facing behaviour for surfaces that are **not implemented**. The nine screens are design evidence only. Nothing in this subsection may be read as delivered capability, and visual approval of a design does not constitute functional acceptance.

Shall. The system shall present the executive user a stable KPI band whose every displayed value resolves to a governed aggregate result and carries an accessible route to its source.

Trace. Source BRD: FR-07, BR-02, FR-15. Process: P-03. Use case: UC-03. UI trace: Executive Insight (Primary 01); Full KPI Dashboard (Secondary 01). Test trace: functional route test; UAT-10; accessibility audit. Dependencies: DEP-05, DEP-07.

Actor / component. Executive Insight surface over the application API.

Preconditions. Authenticated viewer session; accepted dataset generation. **Inputs.** Session role and selected period. **Outputs.** KPI values with lineage affordances.

Required behaviour. KPI composition is stable across sessions for the same generation and period. Values render in approved AED formatting with separators and percentages to one decimal.

Alternative / error behaviour. Loading, empty, error and denied states are explicit and accessible; no KPI tile displays a stale or placeholder value.

Security and access rule. Aggregate-only per SFR-013. **Audit and evidence.** Surface load, role and period selection are recorded.

Acceptance method. Functional route test; accessibility audit under NFR-15; UAT-10 at business acceptance.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Mock-up values are illustrative under I-05 and must not be hardcoded; minimum supported viewport is unapproved under NFR-16.

SFR-018 — Executive Insight governed narrative and chart canvas

Shall. The system shall present governed narrative and charts in the executive canvas, where every figure has passed grounding validation and every chart derives from a session result under SFR-015.

Trace. Source BRD: FR-07, FR-06, FR-04. Process: P-03. Use case: UC-02, UC-03. UI trace: Executive Insight. Test trace: functional route test; UAT-10. Dependencies: DEP-05, DEP-07.

Actor / component. Executive Insight surface; response assembler.

Preconditions. Authenticated viewer session. **Inputs.** Natural-language question. **Outputs.** Narrative, optional chart, and citations.

Required behaviour. Analyst and AI-generated content are visually distinct. Citations resolve to query identifiers.

Alternative / error behaviour. The grounding-blocked state under SFR-011 renders as a deliberate accessible panel and never displays the withheld figure.

Security and access rule. Aggregate-only per SFR-013. **Audit and evidence.** Question, tool calls and response are recorded.

Acceptance method. Functional route test; UAT-10.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. None beyond DEP-05.

SFR-019 — Executive Insight source affordance

Shall. The system shall provide, for every displayed figure and chart on the executive surface, an affordance that opens the source view, query identifier, parameters, execution time and configuration lineage.

Trace. Source BRD: FR-07, FR-12, BR-05. Process: P-03. Use case: UC-03, UC-08. UI trace: Source and Evidence Detail (Secondary 05). Test trace: functional route test; UAT-10. Dependencies: DEP-05.

Actor / component. Executive Insight surface; evidence view.

Preconditions. A displayed governed value. **Inputs.** User activation of the affordance. **Outputs.** Resolved lineage detail.

Required behaviour. Lineage opens without re-executing the query, so that the presented lineage describes the figure actually shown.

Alternative / error behaviour. An unresolvable lineage reference presents an explicit error rather than a blank panel.

Security and access rule. The evidence view exposes no raw secrets and no member identifiers to the viewer role. **Audit and evidence.** Evidence access is itself an audited event.

Acceptance method. Functional route test; reconciliation test against backend results.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Data classification of lineage fields is unresolved under I-06.

8.10 Analyst Workbench surface

SFR-020 — Analyst Workbench governed query surface

Shall. The system shall provide the analyst a governed question surface returning record-scoped results within the analyst access boundary, with citations resolving to query identifiers.

Trace. Source BRD: FR-08, BR-03, FR-01. Process: P-01, P-04. Use case: UC-04, UC-05. UI trace: Analyst Workbench (Primary 02). Test trace: functional route test; role-negative test; UAT-03. Dependencies: DEP-01, DEP-03, DEP-05, DEP-07.

Actor / component. Analyst Workbench surface over the application API.

Preconditions. Authenticated analyst session. **Inputs.** Natural-language question. **Outputs.** Governed results with citations.

Required behaviour. Results respect the analyst boundary; admin-only functions remain unavailable.

Alternative / error behaviour. Empty, error, denied and grounding-blocked states are explicit and accessible.

Security and access rule. Record-level access is permitted within the analyst scope and purpose. **Audit and evidence.** Question, tool calls, parameters and response are recorded.

Acceptance method. Functional route test; role-negative test; UAT-03.

Status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented** · MoSCoW (source): Must.

Open items. Purpose-limitation semantics for record-level access are unresolved under I-06 and SEC-10.

SFR-021 — Analyst Workbench denial-queue surface

Shall. The system shall present the recoverable-denial queue to the analyst with per-item ranking, deadline, evidence-readiness and payer attributes reconciling to the governed result under SFR-007 to SFR-009.

Trace. Source BRD: FR-08, FR-16, FR-03, BR-03. Process: P-01. Use case: UC-04. UI trace: Analyst Workbench; Recoverable Denials Queue (Secondary 02). Test trace: functional route test; reconciliation test; UAT-11. Dependencies: DEP-03, DEP-05.

Actor / component. Analyst Workbench queue component.

Preconditions. Authenticated analyst session; an eligible denial set. **Inputs.** Filter and sort selections. **Outputs.** An ordered, filterable queue.

Required behaviour. Displayed order matches the governed ranking; no client-side reordering alters the recorded priority basis.

Alternative / error behaviour. An empty eligible set renders an explicit empty state distinguishable from a load failure.

Security and access rule. Queue is unavailable to the viewer role. **Audit and evidence.** Filter selections and item selections are recorded.

Acceptance method. Functional route test; reconciliation of every displayed field to the source result; UAT-11.

Status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented** · MoSCoW (source): Must.

Open items. Evidence-readiness definition is unresolved under I-06.

SFR-022 — Analyst Workbench appeal-review surface

Shall. The system shall present an appeal draft to the analyst in an explicitly pending-review state, with the evidence checklist, the governed claim particulars and the review actions available to the analyst role.

Trace. Source BRD: FR-08, FR-10, BR-04. Process: P-02. Use case: UC-06, UC-07. UI trace: Analyst Workbench; Full Appeal Letter (Secondary 04). Test trace: functional route test; UAT-06, UAT-11. Dependencies: DEP-01, DEP-05, DEP-07.

Actor / component. Analyst Workbench appeal component.

Preconditions. An existing draft under SFR-026. **Inputs.** Analyst review actions. **Outputs.** An edited draft that remains pending review.

Required behaviour. The pending-review label is not removable from within this surface. No send control exists anywhere on it.

Alternative / error behaviour. Where required evidence is incomplete, the surface reports the gap and does not offer progression.

Security and access rule. The analyst may prepare and return a draft; the analyst role does not itself carry approval authority, which remains unassigned under D-BRD-07. **Audit and evidence.** Every edit and review action is recorded against the draft.

Acceptance method. Functional route test; UAT-06.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. The approving role is unresolved under D-BRD-07; the letter body is a stub under R-06.

8.11 Audit Console surface

SFR-023 — Audit Console event search and retrieval

Shall. The system shall allow an authorised administrator to search audit events and retrieve, for each, the user, role, time, prompt, tool calls and parameters, result hashes, rules, model and

configuration versions, the response and any approval event.

Trace. Source BRD: FR-09, NFR-05, SEC-07, BR-05. Process: P-04. Use case: UC-08. UI trace: Audit Console (Primary 03). Test trace: functional route test; UAT-08, UAT-12. Dependencies: DEP-05, DEP-06, DEP-07.

Actor / component. Audit Console; persistent audit service.

Preconditions. Authenticated admin session; an operating persistent audit service. **Inputs.** Search criteria. **Outputs.** Matching events with complete fields.

Required behaviour. Retrieved fields reconcile mutually; an event referencing a result hash resolves to the retained result.

Alternative / error behaviour. An unresolvable reference is reported explicitly rather than rendered blank.

Security and access rule. Console access is admin-only and is itself audited. **Audit and evidence.** Audit access events are recorded.

Acceptance method. Functional route test; UAT-08 and UAT-12.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must. No persistent application audit service exists.

Open items. Retention schedule is unapproved under NFR-14 and DEP-06.

SFR-024 — Rules, model and configuration version history

Shall. The system shall present the history of rules, model and configuration versions, and shall associate each historical result with the versions in force when it was produced.

Trace. Source BRD: FR-09, FR-12, NFR-02. Process: P-04, P-05. Use case: UC-08. UI trace: Audit Console version history. Test trace: functional route test; UAT-12. Dependencies: DEP-05, DEP-06.

Actor / component. Audit Console; model and rules registry; configuration store.

Preconditions. Retained version records. **Inputs.** Version or date selection. **Outputs.** Version history with associated results.

Required behaviour. A superseded version remains retrievable; superseding does not delete the prior record.

Alternative / error behaviour. A gap in version history is reported as a gap rather than interpolated.

Security and access rule. Admin-only. **Audit and evidence.** Version transitions are themselves recorded events.

Acceptance method. Functional route test; UAT-12.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must. Configuration version and hash stamping exists in the backend; the presented history does not.

Open items. None beyond DEP-05.

SFR-025 — Append-only audit enforcement

Shall. The system shall record audit events append-only, and shall reject every attempt to alter or delete a recorded event.

Trace. Source BRD: FR-09, NFR-05, SEC-07. Process: P-04. Use case: UC-08. UI trace: Audit Console. Test trace: automated negative test; UAT-12. Dependencies: DEP-05, DEP-06.

Actor / component. Persistent audit service.

Preconditions. An operating audit store. **Inputs.** Alteration or deletion attempt. **Outputs.** Rejection, and a recorded attempt.

Required behaviour. Rejection applies to privileged operators as well as application paths; a correction is recorded as a new event referencing the original.

Alternative / error behaviour. Audit-write failure is itself monitored and alerts, per the operational model.

Security and access rule. No role carries audit-alteration capability. **Audit and evidence.** Rejected attempts are retained as negative evidence.

Acceptance method. Automated negative test with retained evidence; UAT-12.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. The audit service is undelivered under DEP-05; retention and legal hold are unapproved under NFR-14.

8.12 Appeal preparation and human approval

SFR-026 — Governed appeal draft generation

Shall. The system shall generate an appeal draft from governed claim and evidence data, and shall derive every claim particular, amount and date in the draft from a governed tool result rather than from model generation.

Trace. Source BRD: FR-10, BR-04, BR-03. Process: P-02. Use case: UC-06. UI trace: Full Appeal Letter. Test trace: automated unit test; UAT-06. Dependencies: DEP-01, DEP-02, DEP-03, DEP-05, DEP-07.

Actor / component. `draft_appeal` tool; response assembler.

Preconditions. A denied, in-window claim; analyst role. **Inputs.** Claim identifier and tone selection. **Outputs.** An editable draft in pending-review state.

Required behaviour. Claim particulars and the filing window are governed values. The narrative body is generated within the approved template boundary and remains subject to SFR-028.

Alternative / error behaviour. A request against a non-denied or out-of-window claim is rejected and no draft is produced.

Security and access rule. Draft generation is unavailable to the viewer role. **Audit and evidence.** Draft creation, source claim and governed inputs are recorded.

Acceptance method. Automated unit test; UAT-06.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must. The current letter body is a verified stub; claim particulars and window are real.

Open items. The stub must not be presented or used as a production function under R-06.

SFR-027 — Pending-review state; no autonomous approval or send

Shall. The system shall mark every appeal draft pending review, and shall provide no path by which any model, tool, automation or scheduled process approves a draft or transmits it to any external recipient.

Trace. Source BRD: FR-10, BR-04, SEC-08, D-BRD-05. Process: P-02. Use case: UC-06, UC-07. UI trace: Full Appeal Letter; Analyst Workbench. Test trace: automated negative test; role-negative test; UAT-06. Dependencies: DEP-01, DEP-05.

Actor / component. Appeal tool boundary; application API when implemented.

Preconditions. An existing draft. **Inputs.** Any request to approve or send. **Outputs.** Refusal; the draft remains pending review.

Required behaviour. The prohibition is absolute and is not relaxable by configuration, role, prompt or urgency. External submission remains outside the P5 boundary unless separately approved.

Alternative / error behaviour. An approval attempt from a non-human path is refused and recorded as a control event.

Security and access rule. Enforced in code, not in prompt text. **Audit and evidence.** Refused automated approval or send attempts are retained as negative control evidence.

Acceptance method. Automated negative test proving no tool or model path can approve or send; UAT-06.

Status. Catalogue: **Implemented** (safety boundary) · RTM: **Designed / not implemented** (workflow) · MoSCoW (source): Must. The prohibition is implemented at the tool boundary; the named-human workflow it protects is not.

Open items. The authorised approving role is unresolved under D-BRD-07.

SFR-028 — Prohibited-language and evidence-checklist validation

Shall. The system shall validate an appeal draft against the approved prohibited-language list and the required evidence checklist, and shall report every failure before the draft is offered for human review.

Trace. Source BRD: FR-10, SEC-09. Process: P-02. Use case: UC-06. UI trace: Full Appeal Letter. Test trace: automated unit test; UAT-06. Dependencies: DEP-02, DEP-05.

Actor / component. Appeal validation component.

Preconditions. A generated draft. **Inputs.** Draft text and attached evidence references.

Outputs. A pass result or an itemised failure list.

Required behaviour. The draft shall contain no clinical, coverage, eligibility, fraud or misconduct determination beyond the approved deterministic rules.

Alternative / error behaviour. A failing draft remains available for editing and is clearly marked as failing; it is not silently discarded.

Security and access rule. Validation results are visible to the analyst and to the approving role.

Audit and evidence. Validation outcome is recorded with the draft version.

Acceptance method. Automated unit test with planted prohibited language; UAT-06.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. The approved prohibited-language list requires compliance validation under DEP-02.

SFR-029 — Named-human approval event recording

Shall. The system shall record, for every appeal approval, rejection or return, the named human identity, the role, the time, the resource and the decision, as a single immutable event.

Trace. Source BRD: FR-11, BR-04, SEC-08, NFR-05. Process: P-02. Use case: UC-07. UI trace: Full Appeal Letter; Audit Console. Test trace: functional route test; UAT-06, UAT-08. Dependencies: DEP-01, DEP-05, DEP-07.

Actor / component. Approval endpoint; persistent audit service.

Preconditions. An authenticated named human holding the approving role. **Inputs.** Decision and optional comment. **Outputs.** An immutable approval event.

Required behaviour. Identity is the authenticated named identity, never a shared or service account.

Alternative / error behaviour. An approval attempt by an unauthenticated or unauthorised identity is refused and recorded.

Security and access rule. The approving role is distinct from the drafting role. **Audit and evidence.** The event is the primary control evidence for BR-04.

Acceptance method. Functional route test; UAT-06 and UAT-08.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. The role authorised to approve is unassigned under D-BRD-07; authentication is undelivered under DEP-05.

SFR-030 — Duplicate approval prevention and immutable retrieval

Shall. The system shall record exactly one approval decision per resource per decision cycle, shall block a duplicate submission, and shall make the recorded event retrievable without alteration.

Trace. Source BRD: FR-11, NFR-05, SEC-07. Process: P-02. Use case: UC-07, UC-08. UI trace: Audit Console. Test trace: automated negative test; UAT-08. Dependencies: DEP-05.

Actor / component. Approval endpoint; audit service.

Preconditions. An existing approval event. **Inputs.** A repeat submission. **Outputs.** Rejection of the duplicate; the original remains authoritative.

Required behaviour. Idempotency is enforced server-side by resource and decision cycle, not by interface control alone.

Alternative / error behaviour. A genuine subsequent decision on a returned draft opens a new decision cycle rather than overwriting the prior event.

Security and access rule. No role may alter a recorded decision. **Audit and evidence.** Both the original and the blocked duplicate are retained.

Acceptance method. Automated negative test; UAT-08.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Must.

Open items. Decision-cycle semantics require confirmation alongside D-BRD-07.

8.13 Evidence and lineage

SFR-031 — End-to-end evidence lineage resolution

Shall. The system shall resolve, for any generated figure or artefact, the source view, query identifier, parameters, execution time, configuration version and configuration hash, and shall reconcile those to the displayed output.

Trace. Source BRD: FR-12, BR-05, NFR-01. Process: P-03, P-04, P-05. Use case: UC-05, UC-08. UI trace: Source and Evidence Detail. Test trace: reconciliation test; UAT-08. Dependencies: DEP-03, DEP-05, DEP-07.

Actor / component. Evidence and lineage store; evidence view.

Preconditions. A stamped result envelope under SFR-003. **Inputs.** A displayed figure or its identifier. **Outputs.** Complete lineage.

Required behaviour. Reconciliation is exact: the lineage-resolved value equals the displayed value.

Alternative / error behaviour. A mismatch between lineage and display is treated as a defect and the figure is withheld.

Security and access rule. The evidence view exposes no raw secrets and no personal identifiers beyond the requesting role's scope. **Audit and evidence.** Evidence retrieval is audited.

Acceptance method. Reconciliation test against backend results; UAT-08.

Status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented** · MoSCoW (source): Must. Backend lineage elements exist; end-to-end presentation does not.

Open items. Classification of lineage fields is unresolved under I-06.

8.14 Governed data onboarding

SFR-032 — Extract schema, enum and window validation

Shall. The system shall validate a candidate extract against the configured schema, the configured enumerations and a non-overlapping period window before any append occurs.

Trace. Source BRD: FR-13, BR-06. Process: P-05. Use case: UC-01. Test trace: automated unit test; mutation-swap test; UAT-09. Dependencies: DEP-03, DEP-08.

Actor / component. Configuration-driven onboarding pipeline.

Preconditions. A dataset configuration and a candidate extract. **Inputs.** Extract and configuration reference. **Outputs.** Validation pass or an itemised rejection.

Required behaviour. Validation is configuration-driven; table roles, grain, relationships and column meanings are never inferred.

Alternative / error behaviour. A period overlap with an accepted generation is rejected; no partial append occurs.

Security and access rule. Onboarding runs under ingestion credentials separated from the agent plane per SFR-037. **Audit and evidence.** Validation outcome and rejection reasons are recorded in the decision log.

Acceptance method. Automated unit test; mutation-swap end-to-end test; UAT-09.

Status. Catalogue: **Implemented** · RTM: **Implemented** · MoSCoW (source): Must. Both sources agree.

Open items. Approved extracts, mappings and quality thresholds are outstanding under DEP-03.

SFR-033 — Generation tagging and artefact/truth regeneration

Shall. The system shall append a validated extract under a generation tag, advance the governed as-of date, and regenerate dependent artefacts and golden truths against the new generation.

Trace. Source BRD: FR-13, BR-06, NFR-02. Process: P-05. Use case: UC-01. Test trace: automated unit test; regeneration diff; UAT-09. Dependencies: DEP-03.

Actor / component. Onboarding pipeline; artefact generators.

Preconditions. A validated extract. **Inputs.** Extract and configuration. **Outputs.** Tagged generation; regenerated artefacts and truths; a diff report.

Required behaviour. Regeneration is byte-deterministic for a given configuration, and the diff is always produced, including when empty.

Alternative / error behaviour. A regeneration failure aborts before acceptance and leaves the prior accepted state in force.

Security and access rule. Ingestion credentials only. **Audit and evidence.** Generation tag, configuration version and hash, and the diff report are retained.

Acceptance method. Automated unit test; empty-diff regression gate; UAT-09.

Status. Catalogue: **Implemented** · RTM: **Implemented** · MoSCoW (source): Must.

Open items. None specific to this requirement.

SFR-034 — Dry-run evaluation gate and human acceptance decision

Shall. The system shall execute a dry evaluation gate over the regenerated state and shall require an explicit human accept or reject decision before the generation becomes the accepted operating baseline.

Trace. Source BRD: FR-13, BR-06, SEC-08. Process: P-05. Use case: UC-01. Test trace: automated unit test; UAT-09. Dependencies: DEP-03, DEP-07, DEP-08.

Actor / component. Onboarding pipeline; evaluation harness; Data Owner/Steward.

Preconditions. A regenerated candidate state. **Inputs.** Gate results and the human decision.

Outputs. Acceptance or rejection, recorded.

Required behaviour. No generation is accepted automatically on a passing gate; the human decision is mandatory and named.

Alternative / error behaviour. A failing gate blocks acceptance and offers only rejection and rollback.

Security and access rule. The acceptance decision is restricted to the Data Owner/Steward role. **Audit and evidence.** Gate results and the named decision are retained.

Acceptance method. Automated unit test; UAT-09.

Status. Catalogue: **Implemented** · RTM: **Implemented** · MoSCoW (source): Must.

Open items. Representative-data assumption A-06 remains open.

SFR-035 — Atomic rollback on generation rejection

Shall. On rejection of a generation the system shall restore data, configuration, artefacts and golden truths to the prior accepted state, and the restored checksums shall match that prior state.

Trace. Source BRD: FR-13, NFR-06, BR-06. Process: P-05. Use case: UC-01. Test trace: automated rollback test; checksum comparison; UAT-09. Dependencies: DEP-03, DEP-08.

Actor / component. Onboarding pipeline rollback path.

Preconditions. A rejected generation. **Inputs.** Rejection decision. **Outputs.** Restored prior state; retained decision log.

Required behaviour. Rollback is atomic across all four component classes; no mixed-version state is reachable.

Alternative / error behaviour. A rollback failure raises an operational alert and blocks further onboarding until resolved.

Security and access rule. Ingestion credentials only. **Audit and evidence.** The rejection, the rollback and the resulting checksums are retained.

Acceptance method. Automated rollback test with checksum comparison; UAT-09.

Status. Catalogue: **Implemented** · RTM: **Implemented** · MoSCoW (source): Must.

Open items. None specific to this requirement.

8.15 Agent-plane boundary

SFR-036 — Read-only agent-plane boundary

Shall. The system shall expose source and replica data to the agent plane as read-only, and shall deny every agent-plane attempt to insert, update or delete data.

Trace. Source BRD: FR-14, SEC-10, NFR-03. Process: P-03, P-04, P-05. Use case: UC-01, UC-02, UC-04. Test trace: automated negative test. Dependencies: DEP-03, DEP-05.

Actor / component. Semantic layer; replica access boundary.

Preconditions. An active agent session. **Inputs.** Any mutating operation. **Outputs.** Denial.

Required behaviour. The boundary is a property of the access path, not a convention observed by tool authors.

Alternative / error behaviour. Denied mutation attempts are recorded and do not degrade read capability.

Security and access rule. Least privilege; separation of duties from ingestion. **Audit and evidence.** Retained tests demonstrate the boundary.

Acceptance method. Automated negative test proving the agent role cannot mutate source or replica.

Status. Catalogue: **Implemented** · RTM: **Implemented** · MoSCoW (source): Must.

Open items. None specific to this requirement.

Shall. The system shall hold ingestion credentials separately from agent-plane credentials, and no single credential shall carry both ingestion write capability and agent query capability.

Trace. Source BRD: FR-14, SEC-10, SEC-04. Process: P-05. Use case: UC-01. Test trace: inspection; automated negative test. Dependencies: DEP-03, DEP-05.

Actor / component. Onboarding pipeline; agent dispatcher; credential store.

Preconditions. Configured credential separation. **Inputs.** Credential resolution at execution.

Outputs. Scoped access.

Required behaviour. Separation is enforced at the credential level rather than by process convention.

Alternative / error behaviour. A combined-capability credential is a control failure and blocks onboarding.

Security and access rule. Secrets are held in an approved secret manager and never in source, chat, screenshot or log. **Audit and evidence.** Credential scope is evidenced by inspection and retained tests.

Acceptance method. Inspection plus automated negative test.

Status. Catalogue: **Implemented** · RTM: **Implemented** · MoSCoW (source): Must. The architectural and code boundary is verified; production secret management is not delivered.

Open items. Production IAM and secret management remain outstanding under DEP-05 and NFR-03.

8.16 Should-priority requirements

📌 SFR-038 to SFR-042 inherit **Should** priority from MoSCoW Pass #1 (FR-15 to FR-18). They are included in the intended pre-production baseline. Inheriting Should does not reduce their specification rigour and does not permit their acceptance criteria to be waived without change control.

Shall. The system shall provide a full KPI dashboard supporting approved filters, where every filtered value recomputes from a governed result and no filter combination produces an ungrounded figure.

Trace. Source BRD: FR-15, BR-02. Process: P-03. Use case: UC-03. UI trace: Full KPI Dashboard (Secondary 01). Test trace: functional route test; UAT-10. Dependencies: DEP-05,

DEP-07.

Actor / component. Full KPI Dashboard surface.

Preconditions. Authenticated viewer session. **Inputs.** Filter selections. **Outputs.** Filtered governed aggregates.

Required behaviour. Filters operate server-side against governed views; client-side filtering of a retrieved superset is not permitted for privacy-scoped data.

Alternative / error behaviour. A filter combination yielding no data renders an explicit empty state.

Security and access rule. Aggregate-only per SFR-013. **Audit and evidence.** Filter selections are recorded.

Acceptance method. Functional route test; UAT-10.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Should.

Open items. Approved filter set is design evidence only.

SFR-039 — Export controls and approved formatting

Shall. The system shall apply role, purpose and classification controls to every export, and shall render exported values in the approved formatting convention.

Trace. Source BRD: FR-15, SEC-05, SEC-10. Process: P-03, P-01. Use case: UC-03, UC-04. UI trace: export affordances. Test trace: role-negative test; functional route test. Dependencies: DEP-05, DEP-07.

Actor / component. Export component; application API.

Preconditions. An authorised role and a stated purpose. **Inputs.** Export request. **Outputs.** A scoped export artefact.

Required behaviour. Viewer-role exports contain aggregates only. AED values carry separators; percentages carry one decimal; British spelling applies.

Alternative / error behaviour. An export exceeding the role scope is denied rather than silently truncated.

Security and access rule. Abnormal export activity is monitored per the operational model.

Audit and evidence. Export requests, scope and outcome are recorded.

Acceptance method. Role-negative test across export paths; functional route test.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Should.

Open items. Purpose-limitation and classification semantics are unresolved under I-06 and SEC-10.

SFR-040 — Full recoverable-denials queue behaviour

Shall. The system shall provide a full recoverable-denials queue supporting search, filter, priority ordering, evidence-readiness indication and deadline display, with every field reconciling to governed results.

Trace. Source BRD: FR-16, FR-03, BR-03. Process: P-01. Use case: UC-04. UI trace: Recoverable Denials Queue (Secondary 02). Test trace: functional route test; reconciliation test; UAT-11. Dependencies: DEP-03, DEP-05.

Actor / component. Recoverable Denials Queue surface.

Preconditions. Authenticated analyst session. **Inputs.** Search and filter criteria. **Outputs.** A governed queue view.

Required behaviour. Search and filter operate against governed results; the displayed priority basis is the governed ranking under SFR-008.

Alternative / error behaviour. Empty and error states are distinguishable; no required action is blocked at the approved viewports.

Security and access rule. Analyst scope; unavailable to viewer. **Audit and evidence.** Search and selection events are recorded.

Acceptance method. Functional route test; reconciliation test; UAT-11.

Status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented** · MoSCoW (source): Should. The backend queue feed exists; the queue surface does not.

Open items. Evidence-readiness definition unresolved under I-06.

SFR-041 — Insights library with severity, impact and drill-down

Shall. The system shall provide an insights library in which every stored finding carries severity, quantified impact, resolvable citations and a drill-down path to its evidence.

Trace. Source BRD: FR-17, BR-05. Process: P-03. Use case: UC-03. UI trace: All Insights (Secondary 03). Test trace: functional route test; reconciliation test; UAT-10. Dependencies: DEP-05.

Actor / component. All Insights surface; insight store.

Preconditions. Stored findings produced from governed results. **Inputs.** Library navigation and filters. **Outputs.** Findings with resolvable citations.

Required behaviour. A stored finding retains the lineage of the result that produced it; a finding whose citation no longer resolves is marked stale rather than displayed as current.

Alternative / error behaviour. An empty library renders an explicit empty state.

Security and access rule. Findings inherit the access scope of their underlying results. **Audit and evidence.** Finding creation and access are recorded.

Acceptance method. Functional route test; citation-resolution test; UAT-10.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Should.

Open items. Stale-finding handling requires confirmation as part of section 10.

SFR-042 — Profile, preferences and access presentation

Shall. The system shall present the authenticated user their identity, role, effective permissions and notification preferences, and the displayed permissions shall match server-side enforcement exactly.

Trace. Source BRD: FR-18, SEC-04, NFR-03. Process: P-03. Use case: UC-09. UI trace: My Profile (Secondary 06). Test trace: functional route test; reconciliation against the role matrix. Dependencies: DEP-05, DEP-07.

Actor / component. My Profile surface; identity and access provider dependency.

Preconditions. An authenticated session. **Inputs.** Profile navigation. **Outputs.** Identity, role, permissions and preferences.

Required behaviour. Displayed permissions are read from the enforcement source, not from a duplicated presentation list that could drift.

Alternative / error behaviour. Where the identity provider is unavailable, the surface reports the degraded state and does not display a cached permission set as current.

Security and access rule. The profile discloses the user's own scope only. **Audit and evidence.** Profile access and preference changes are recorded.

Acceptance method. Functional route test; reconciliation of displayed permissions against the enforced role matrix.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Should.

Open items. SSO and session evidence require the identity provider dependency under DEP-05.

8.17 Future Gen-3 requirements — segregated, not in the minimum baseline

⚠ SFR-043 to SFR-046 derive from BRD FR-19 to FR-22, which hold **Could** priority and are **future Gen-3 capability**. They are recorded here for completeness and traceability only. They are not part of the minimum pre-production baseline, they carry no delivery commitment, and they require separate approval and implementation. Gen-3 documents are design evidence and may overstate implementation; repository verification confirms no Gen-3 capability exists. These requirements are specified at outline level deliberately — elaborating them to delivery depth would misrepresent their governance state.

SFR-043 — Adaptive single-table intake profiling (future Gen-3)

Shall. Where separately approved and implemented, the system shall profile an unfamiliar single-table RCM extract and propose table role, grain and column mappings for explicit human confirmation, and shall not adopt any inferred mapping without that confirmation.

Trace. Source BRD: FR-19. Process: P-05. Use case: extends UC-01. Test trace: to be defined at Gen-3 approval. Dependencies: separate Gen-3 approval; DEP-03.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Could. **Open items.** The current onboarding boundary in SFR-032 explicitly forbids inference, and that boundary governs until Gen-3 is separately approved. Recorded in RAID I-04.

SFR-044 — Period comparison and change decomposition (future Gen-3)

Shall. Where separately approved and implemented, the system shall compare periods and decompose changes using server-computed values only, with hedged narration and no model-derived arithmetic.

Trace. Source BRD: FR-20. Process: P-03. Use case: extends UC-02, UC-03. Test trace: to be defined at Gen-3 approval. Dependencies: separate Gen-3 approval.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Could. **Open items.** The grounding boundary in SFR-010 applies unchanged to any future Gen-3 capability.

SFR-045 — Deterministic anomaly detection (future Gen-3)

Shall. Where separately approved and implemented, the system shall detect anomalies using versioned deterministic methods verified by planted-anomaly tests.

Trace. Source BRD: FR-21. Process: P-03. Use case: extends UC-03. Test trace: planted-anomaly tests, to be defined at Gen-3 approval. Dependencies: separate Gen-3 approval.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Could. **Open items.** None; not in baseline.

SFR-046 — Proactive cited findings feed (future Gen-3)

Shall. Where separately approved and implemented, the system shall file proactive findings with resolvable citations under a least-privilege monitor identity, and that identity shall carry no approval or outbound capability.

Trace. Source BRD: FR-22, SEC-08. Process: P-03. Use case: extends UC-03. Test trace: to be defined at Gen-3 approval. Dependencies: separate Gen-3 approval.

Status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented** · MoSCoW (source): Could. **Open items.** The prohibition in SFR-027 applies to any monitor identity without exception.

8.18 Deferred capability — exclusion requirements

✖ SFR-047 to SFR-050 correspond to BRD FR-23 to FR-26, priority **Won't in this baseline**, status **Deferred** under decision D-BRD-05. They are stated as negative system requirements because an exclusion that is testable is a control, whereas an exclusion that is only narrated is an assumption. **These must not become active delivery stories.** Any future change requires a new approved scope and design package under Change Control.

SFR-047 — Exclusion: no autonomous external appeal submission

Shall not. The system shall not transmit an appeal to any payer, TPA or external submission channel by any automated path. External submission remains outside the P5 boundary.

Trace. Source BRD: FR-23, BR-04, SEC-08, D-BRD-05. Use case: boundary of UC-07. Test trace: automated negative test. **Acceptance method.** Automated negative test proving no submission path exists. **Status.** Deferred / Deferred · MoSCoW (source): Won't. **Open items.** Any future external submission requires a separate approved scope package and does not inherit this baseline's authority.

SFR-048 — Exclusion: no production claims or EHR write

Shall not. The system shall not write to any production claims, EHR, clearinghouse, payment or member-contact system.

Trace. Source BRD: FR-24, SEC-10, D-BRD-05. Use case: boundary of all use cases. Test trace: automated negative test; inspection of credential scope. **Acceptance method.** Automated

negative test plus credential-scope inspection under SFR-037. **Status.** Deferred / Deferred · MoSCoW (source): Won't. **Open items.** None.

SFR-049 — Exclusion: no automatic multi-table relationship discovery

Shall not. The system shall not infer relationships across multiple tables automatically. Onboarding remains configuration-driven per SFR-032.

Trace. Source BRD: FR-25. Use case: boundary of UC-01. Test trace: inspection; onboarding contract test. **Acceptance method.** Inspection of the onboarding contract; automated test that no relationship inference path is reachable. **Status.** Deferred / Deferred · MoSCoW (source): Won't. **Open items.** Gen-3 Tier 2 scope; recorded in RAID I-04.

SFR-050 — Exclusion: no autonomous workers or scheduled cohort scoring

Shall not. The system shall not run autonomous workers, scheduled cohort scoring or unattended production scoring pipelines.

Trace. Source BRD: FR-26, SEC-08, D-BRD-05. Use case: boundary of all use cases. Test trace: automated negative test; inspection of scheduling configuration. **Acceptance method.** Inspection plus automated negative test. **Status.** Deferred / Deferred · MoSCoW (source): Won't. **Open items.** `score_member` is a verified stub and must not be presented as production scoring, per R-06.

8.19 Coverage statement

Every BRD functional requirement FR-01 to FR-26 is represented by at least one SFR. Coverage of the mandated behaviour areas is as follows: typed governed queries SFR-001 to SFR-003; deterministic compliance verdicts SFR-004 to SFR-006; recoverable-denials ranking SFR-007 to SFR-009; numeric grounding validation SFR-010 to SFR-011; role enforcement SFR-012 to SFR-014; chart-contract validation SFR-015 to SFR-016; Executive Insight SFR-017 to SFR-019; Analyst Workbench SFR-020 to SFR-022; Audit Console SFR-023 to SFR-025; governed appeal drafting SFR-026 to SFR-028; named-human approval recording SFR-029 to SFR-030; evidence and lineage SFR-003 and SFR-031; atomic onboarding and rollback SFR-032 to SFR-035; read-only agent-plane boundary SFR-036 to SFR-037; KPI, filter and export behaviour SFR-038 to SFR-039; recoverable-denials queue SFR-040; insights library SFR-041; profile and access presentation SFR-042; future Gen-3 capability SFR-043 to SFR-046; deferred autonomous and write capability SFR-047 to SFR-050.

No SFR asserts an implementation status beyond what the parent document's verified-evidence section confirms. No numerical target is introduced in this section; all such targets belong to [section 9](#) and are recorded there as approved or as TBD.